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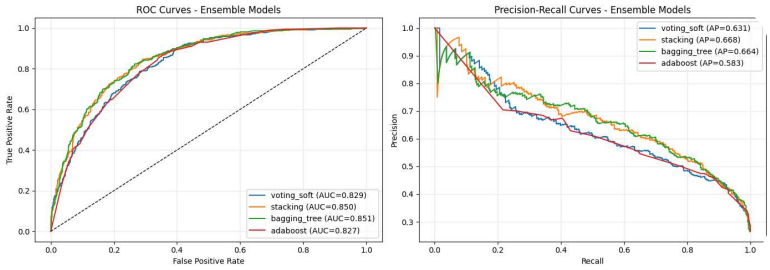

Explain

This figure compares different **ensemble models** using two important evaluation tools: the **ROC curve** and the **Precision–Recall curve**. Let's break it down clearly.

🔵 1. ROC Curve (Left Side)

The **ROC Curve** shows how well a model separates two classes (e.g., pass/fail, churn/not churn).

Axes:

- **X-axis:** False Positive Rate (FPR)
- **Y-axis:** True Positive Rate (TPR)

Key idea:

- The closer the curve is to the **top-left corner**, the better the model.
- The dashed diagonal line = **random guessing**

Metric:

- **AUC (Area Under Curve)**
  Higher = better

Your Results:

Model	AUC	Interpretation
Bagging Tree	0.851	🟢 Best
Stacking	0.850	🟢 Very close
Voting (Soft)	0.829	🟡 Good

Model	AUC	Interpretation
AdaBoost	0.827	🟡 Slightly lower

👉 **Conclusion (ROC):**

- **Bagging and Stacking** perform best at distinguishing classes overall.
- Differences are small → all models are decent.

🟠 2. Precision–Recall Curve (Right Side)

The **Precision-Recall Curve** is more useful when your dataset is **imbalanced** (very common in real-world problems like churn, fraud, medical diagnosis).

Axes:

- **X-axis:** Recall (how many positives you captured)
- **Y-axis:** Precision (how many predicted positives are correct)

Metric:

- **Average Precision (AP)**
  Higher = better balance of precision and recall

Your Results:

Model	AP	Interpretation
Stacking	0.668	🟢 Best
Bagging Tree	0.664	🟢 Very close
Voting (Soft)	0.631	🟡 Moderate
AdaBoost	0.583	🔴 Lowest

👉 **Conclusion (PR Curve):**

- **Stacking is the best overall** when focusing on correct positive predictions.
- **AdaBoost struggles more**, especially at higher recall.

🧠 Final Interpretation

✅ Best Overall Model:

- **Stacking**
  - Strong ROC (AUC ≈ 0.85)
  - Best Precision-Recall performance

✅ Close Competitor:

- **Bagging Tree**
  - Almost identical ROC
  - Slightly lower PR performance

⚠️ Weaker Models:

- **Voting:** decent but not top
- **AdaBoost:** consistently lowest

🚀 Key Insight (Very Important for Your Research)

- If your problem is **balanced** → **use ROC (AUC)**
- If your problem is **imbalanced** → **trust Precision-Recall more**

👉 So for most real-world ML problems (like churn, student failure prediction):
Stacking = best choice here

If you want, I can now:

- ↪ Help you **write this as a research paper explanation (formal)**
- Or ↪ explain **why stacking outperforms others mathematically**